

LAB 5 (SECJ1023)
PROGRAMMING TECHNIQUE II
SECTION 03, 04 & 06, SEM 2, 2025/2026

1. Type in the following program.

```
1 //Program 5.1
2 #include <iostream>
3 #include <string>;
4 using namespace std;
5
6 int main()
7 {
8     string studentName;
9
10    cout << "Type your name:";
11    cin >> studentName;
12    cout << "Welcome " << studentName;
13
14    return 0;
15 }
```

- a. Compile and run the program. What will be displayed if the user enters the following input:

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- b. Explain why such output is displayed. How can the program be improved?

2. Given the following program segment, answer the given questions:

a.

```
1 //Program 5.2
2 string city = "Kuala Kangsar";
3 int len;
4 len = city.length();
5 cout << "The length of the city is " << len;
```

- i. Identify the appropriate header file for the C++ String function.
- ii. Draw the memory layout of the variable `city`.
- iii. Show the output.

b.

```
1 //Program 5.3
2 string message1 = "Happy";
3 string message2 = "Birthday";
4 cout << message1 << endl;
5 cout << message2 << endl;
6 cout << "Size of message1 :" << message1.length() << endl;
7 cout << "Size of message2 :" << message2.length() << endl;
```

8	message1 = message1 + message2;
9	cout << message1 << endl;

- i. Identify the appropriate header file for the C++ String function.
- ii. Draw the memory layout of the variable message1 and message2.
- iii. Show the output.
- iv. Modify the Program 5.3 above by changing the function length() in line 6 and 7 to function size(). State whether the output of the program remains the same, or change.
- v. Show the difference between the usage of function length() and function size() if any.

c.

1	//Program 5.4
2	string str1, str2, str3;
3	
4	str1 = "My dad ";
5	str2 = "like ";
6	str3 = str1 + str2;
7	
8	cout << str1 << endl << str2 << endl << str3 << endl;
9	strcat(str3, "fishing");
10	cout << str3 << endl;

- i. The code segment in Program 5.4 has an error. Identify and fix the error.
- ii. Show the output after fixing the error.

d.

1	//Program 5.5
2	string firstname;
3	string secondname;
4	cout << "Enter first name:";
5	cin >> firstname;
6	cout << "Enter second name:";
7	cin >> secondname;
8	
9	cout << firstname << endl;
10	cout << secondname << endl;
11	if (firstname < secondname)
12	cout << firstname << " should come first before "
13	<< secondname << endl;
14	else if (firstname > secondname)
15	cout << secondname << " should come first before "
16	<< firstname << endl;
17	else
18	cout << "You entered the same name twice!\n";

Show the output and explain why such output is displayed for the following input:

- i. Enter first name:Azlan
Enter second name:Shazrik
 - ii. Enter first name:Hayat
Enter second name:Hafetz
 - iii. Enter first name:najib
Enter second name:Najib
 - iv. Enter first name:abc
Enter second name:ABC
 - v. Enter first name:Foziyah
Enter second name:Foziyah
3. Determine whether the following program has errors. If there are errors, fix it and then state the output in Table 5.1 for the given input.

```

1 //Program 5.6
2 #include <iostream>
3 #include <string>;
4 using namespace std;
5
6 int main()
7 {
8     string myCar = "Proton";
9     cout << "Enter type of car: ";
10    string userCar = getline();
11
12    if (userCar == "Honda")
13        cout << "You have enter a Honda car.\n";
14    else if(userCar == "Toyota")
15        cout << "Hello Toyota car.\n";
16    else if (userCar != myCar)
17        cout << "Your dream car is.\t" << userCar;
18    else
19        cout << "Oh it is your car \t" << myCar;
20
21    return 0;
22 }
```

Table 5.1

No	Input	Output
a.	Bentley	
b.	Proton	
c.	Honda	
d.	Toyota	

4. For each question, identify whether the statement in the question is TRUE or FALSE. Verify your answer with appropriate reasons.

a. The following code segment will print “8”.

```
1 //Program 5.7
2 string a = "1", b = "4", c = "3";
3 if ( a < b )
4     a += c;
5
6 if ( b == a)
7     a += b;
8 else
9     a += c + b;
10
11 cout << a << endl;
```

b. The following code segment will print “a is less than b”.

```
1 //Program 5.8
2 string a, b, c;
3 a = "ABC";
4 b = "XYZ";
5 c = a + b;
6
7 for (int x=0; x < c.size();x++)
8     cout << c[x];
9 cout << endl;
10
11 if (a < b)
12     cout << "a is less than b\n";
13 else
14     cout << "a is not less than b\n";
```

c. The following code segment will print “False”.

```
1 //Program 5.9
2 string s1 = "Five Oranges";
3 string s2 = "Four Oranges";
4
5 if (s1 < s2)
6     cout << "True" << endl;
7 else
8     cout << "False" << endl;
```

5. Given the following incomplete program segment, answer the given questions. The program asks to copy and convert a C++ string object to a C-string.

```
1 //Program 5.10
2 char str1[20];
3 string str2 = "Real Time";
4
```

5	strcpy (_____); //copy the C++ string
6	//into an array str1
7	cout << str1;

- Identify the appropriate header file for Program 5.10.
- Complete the code in Line 5.
- Show the output.

```
1 //Program 5.11
2 string mySentences = "Hello everyone! ";
3 string varStr="Smile";
4 mySentences.append(varStr);
5 cout << mySentences << endl;
```

```
1 //Program 5.12
2 string mySentences = "Hello everyone! ";
3 string varStr="Smile";
4 mySentences.append(varStr, 2, 3);
5 cout << mySentences << endl;
```

```
1 //Program 5.13
2 string mySymbol;
3 mySymbol.assign(3, 'x');
4 cout << mySymbol << endl;
```

```
1 //Program 5.14
2 string myCode;
3 string varStr = "1234-56-7890";
4 myCode.assign(varStr,5,2);
5 cout << myCode << endl;
```

```
1 //Program 5.15
2 string yourAddress = "Jalan Tekukur, Taman Impian";
3 string varStr = "an";
4 cout <<yourAddress.find(varStr,5)<< endl;
```

```
1 //Program 5.16
2 string w("Adam ");
3 string x("Hanny likes durian ");
```

```

4  string y(" rambutan ");
5  string z;
6
7  if (w<x){
8      z=x.substr(5,6);
9      z+=y;
10     w+=z;
11 }
12 else
13     x+=y;
14     cout << w << endl;
15     cout << x << endl;
16     cout << (x+y).find(z.substr(6,5))<< endl;

```

7. Given a program segment in Program 5.17. Determine the output of the following statement:

```

1  //Program 5.17
2  string str1 = "Pulau ";
3  string str2 = "Langkawi";
4  string str3;

```

- `cout << str2.length();`
 - ```

if (str1 < str2)
 cout << str1 << " smaller than " << str2;
else
 cout << str1 << " bigger than " << str2;

```
  - ```

str3 = str1 + str2;
cout << str3;

```
 - `cout << str3.length();`
 - ```

str3 += " Kedah";
cout << str3 << endl;

```
  - ```

for (int x=0; x < str2.size(); x++)
    cout << str2[x];
cout << endl;

```
8. Given an incomplete program which utilizes C++ string objects as follows. Complete the program by filling in the blanks according to the instructions stated in the comments.
Note: You are not allowed to use C-string library functions to answer this question.

```

1  //Program 5.18
2  #include <iostream>
3  _____(a)_____//Include the appropriate library for
4  _____//C++ string manipulation
5  using namespace std;
6  int main ()
7  {
8      string str1="Hello", str2="World", str3;
9      int len;
10     _____(b)_____ // Copy str1 into str3

```

```

11 cout << "str3 : " << str3 << endl;
12
13 ____ (c) ____ // Append str2 to str3
14 cout << "New str3: " << str3 << endl;
15
16 ____ (d) ____ // Store the new length of str3 into len
17
18 cout << "The new length of str3: " << len << endl;
19
20 ____ (e) ____ // Take the first half of str3 and store
21                // it into str2.
22 cout << "First half of the str3: " << str2 << endl;
23
24 return 0;
25 }

```

The program prints the following output when it runs:

```

str3 : Hello
New str3 : HelloWorld
The new length of str3: 10
First half of the str3: Hello

```

9. Given an incomplete program which utilizes C++ string objects as follows. Complete the program by filling in the blanks according to the instructions stated in the comments. Note: You are not allowed to use C-string library functions to answer this question.

```

1 //Program 5.19
2 #include <iostream>
3 ____ (a) ____ //Include the appropriate library for
4                // C++ string manipulation
5 using namespace std;
6 int main ()
7 {
8     string firstname= "Abu", lastname = "Bakar";
9     string fullname;
10    ____ (b) ____ // Concatenate the firstname &
11                // lastname
12    ____ (c) ____ // Append a single char(,)
13    ____ (d) ____ // Append another string(Dr)
14
15    cout << fullname << endl;
16
17    ____ (e) ____ // Replace a string -(Dr)to(Prof Madya)
18    cout << fullname << endl;
19
20    ____ (f) ____ // Erase a string -(Madya)
21    cout << fullname << endl;
22
23    ____ (g) ____ // Insert a string -(Associate)
24    cout << fullname << endl;

```

25	return 0;
26	}

The program prints the following output when it runs:

Abu Bakar,Dr
Abu Bakar,Prof Madya
Abu Bakar,Prof
Abu Bakar,Associate Prof

10. The following code segment has error. Give explanation and correct them. **Note:** You are not allowed to use C-string library functions to answer this question.

a.

1	//Program 5.20
2	string grade = "A+";
3	if(strcmp(grade,"A+") == 0)
4	cout << "same grade";

b.

1	//Program 5.21
2	string firstname = "Uthman";
3	string lastname = " Affan ";
4	strcat(firstname, lastname);

c.

1	//Program 5.22
2	char c = '\\';
3	string s = 'the backslash!';
4	cout << c << s << endl;

d.

1	//Program 5.23
2	string str1 = "Hello";
3	string str2 = " World! ";
4	strcpy(str1, str2);

11. Write a program that allows the user to enter a telephone number. Create a function named **phoneFormat** that inserts dash (-) and a plus (+) sign at the appropriate locations in a string object containing an unformatted telephone number. As an argument, the function should accept a reference to a string object. Assume the string object contains 11 digits telephone number with a value such as 01912345678. The function should modify the string object, so it contains a formatted telephone number, such as +019-1234-5678. Figure 5.1 is an example run of the program. Note that the text in bold indicates input entered by the user.

```
Enter your handphone no. -> 01912345678
Your formatted handphone no. -> +019-1234-5678
```

Figure 5.1

12. Write a program that will display the output as in Figure 5.2. The program should use appropriate member functions and do the following tasks:
- Assign a string and display the output as in Line 1.
 - Insert additional string “**good**” at a right position to display the output as in Line 2.
 - Replace the portion of string “**is a good sub**” in a sentence of Line 2, by a new string “**re**” to display the output as in Line 3.

```
1  Programming is a subject!
2  Programming is a good subject!
3  Programming reject!
```

Figure 5.2

13. Write a program that has an array of at least 10 string objects which hold students’ name and matric numbers. Make your own string, or use the following string for an array.

```
"Muhamad Husin, A123456"
"Nor Asma, A156782"
"Raja Kumar, A187623"
"Muhamad Ali, A123908"
"Ali Shah, A145234"
"Nor Nadia, A128734"
"Siti Sarah, A120945"
"Siti Kahadijah, A154237"
"Syed Azman, A276456"
"Chong Wei, A156278"
```

Figure 5.3

The program should ask the user to enter a name or partial name to search for in the array. Any entries in the array that match the string entered should be displayed. Figure 5.4 is an example run of the program. Note that the text in bold indicates input entered by the user and the search string is case sensitive.

```
Searching Student's Info
-----
Enter the name or partial name,
you would like to search for: Muhamad

Search results:
Muhamad Husin, A123456
Muhamad Ali, A123908
```

Figure 5.4

14. Write a program that asks the user to enter a book details with an author's name, year published and a book title. The author's name takes a string containing a fullname in the form of firstname and lastname separated from each other by a single space. The program should then arrange the input in the following manner:
 - a. Fullname - The lastname followed by a comma, followed by a firstname and then followed by a space.
 - b. Year of publication - Insert an open bracket “(”, followed by year published and then followed by a close bracket “)”.
 - c. Book's title – Insert a space and followed by a book's title.
 - d. Example output – Jason, Adam (2018) Bitcoin: Cryptocurrency

Figure 5.5 is an example run of the program. Note that the text in bold indicates input entered by the user.

```
Enter author of the book -> Adam Jason
Enter year of published -> 2018
Enter a book title -> Bitcoin: Cryptocurrency

The Formatted Book Reference:
Jason,Adam (2018) Bitcoin: Cryptocurrency
```

Figure 5.5

15. Write a program which is able to perform the addition and subtraction operations on numbers one to five. The user is required to enter the calculation he or she wants to perform in english words with small letters and in a single string, for example “two plus one”, and “one minus three”. The result of the calculation should also be displayed in words, for example, “one minus three is negative two”.

Besides the main function, your program should also define a function that converts a number in word to its integer value, for example, “one” to 1, “two” to 2 and so on. Figure 5.6 shows some example runs of what your program should look like. Note that the bold text in each example indicates the input entered by the user. The assessment criteria are given in Table 1.

Run 1	What is: two plus one Answer: two plus one is three
Run 2	What is: one minus three Answer: one minus three is negative two
Run 3	What is: five minus five Answer: five minus five is zero

Run 4	What is: five plus five Answer: five plus five is ten
-------	---

Figure 5.6: Example runs

Table 1: Assessment Criteria

Item	Criteria	Marks
A	i) The program is able to run.	2
	ii) Using an appropriate structure for the program (e.g. the code is properly indented, all the required header files are included, the main function is properly written, etc.).	2
B	The definition of function that converts a number in word to its integer value.	6
C	Reading the text of operation in a single string from the keyboard.	1
D	Extracting the words of the first and second operands as well as the operator.	6
E	Converting the operands from words to integers, e.g., “five” to 5.	2
F	Performing the arithmetic calculation accordingly, i.e., addition or subtraction .	3
G	Converting the result value from integer to word, e.g. 10 to “ten”.	12
H	Printing the answer onto the screen.	1
Total		35

16.